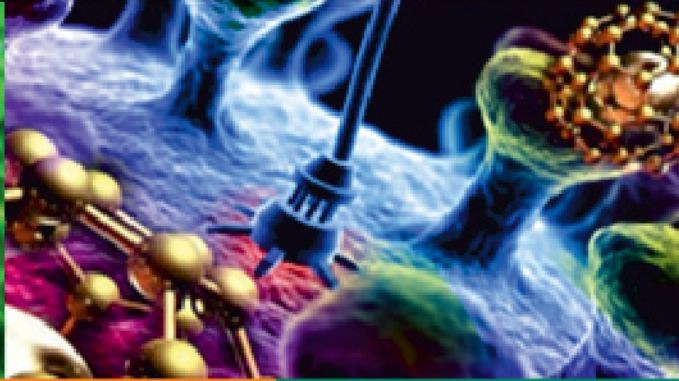
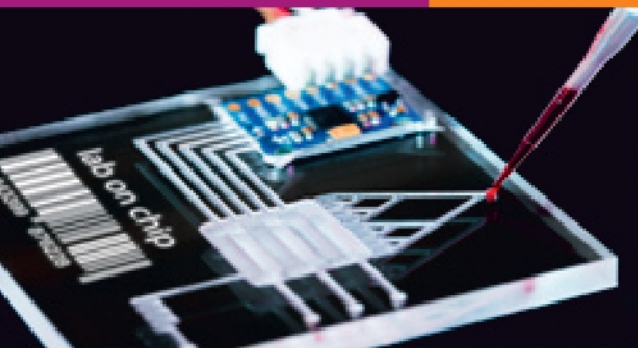


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ABSTRACTS



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PP-015

Aptamer-Tethered Hyperbranched Bis-Mpa Polyester Dendrimers for Targeted Delivery of Sirna and Gene Silencing in Breast Cancer Cells

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Abstract: This study investigates the potential of mucin-1 aptamer-tethered hyper-branched bis-MPA polyester dendrimers as carriers for targeted delivery of Survivin siRNA in breast cancer cells. The objectives included dendrimer synthesis, PEG and mucin-1 aptamer conjugation, and evaluating Survivin gene silencing efficacy in MCF-7 cells. Hydroxyl groups of hyper-branched bis-MPA polyester dendrimers were modified to carboxylic groups, then conjugated with tetraethylenepentamine. Mucin-1 aptamer was attached using heterobifunctional polyethylene glycol as a linker. Characterization via ¹H NMR and dynamic light scattering confirmed successful conjugation steps. Real-time PCR assessed Survivin gene silencing by the designed siRNA-loaded dendrimers. Positive zeta potential and ¹H NMR confirmed tetraethylenepentamine conjugation. Targeted nanoparticles exhibited enhanced serum stability for siRNA and biocompatibility in NIH-3T3 fibroblast cells. Cellular uptake studies in MCF-7 cells revealed efficient internalization of targeted nanoparticles compared to non-targeted ones. Real-time PCR data demonstrated superior inhibition of target gene expression by targeted nanoparticles compared to non-targeted ones and lipofectamine 2000. In conclusion, mucin-1 aptamer-tethered hyper-branched bis-MPA polyester dendrimers effectively silence target genes in breast cancer cells, indicating their potential as promising carriers for precise siRNA delivery in breast cancer therapy.

Keywords: Dendrimer; Mucin-1 aptamer; Breast cancer; siRNA

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